Exc.no:13
Date:

Implement your own ping program.

AIM:

To implement a simple ping program in computer networks to check whether a destination host is reachable and to measure the round-trip time (RTT) of packets.

Algorithm:

- Start the program.
- Read the destination host name or IP address from the user.
- Create a raw ICMP socket.
- Construct an ICMP Echo Request packet containing:
  - ICMP type
  - Code
  - Checksum
  - Identifier
  - sequence number
  - Data.
- Calculate the checksum of the packet.
- Send the ICMP Echo request to the destination.
- Wait for an ICMP Echo Reply.
- Calculate the round-trip time between sending the request and receiving the reply.
- Display the destination, packet size, sequence number, and RTT.
- Repeat the process for a specified number of packets.
- Display the final result and terminate.

output

Enter host name or IP address : 8.8.8.8

Reply from 8.8-8.8 bytes =43 time =24.5 ms

Reply from 8-8-8-8 bytes =43 time =23.87 ms

Reply from 8.8.8.8.bytes = 43 time = 24.12 ms

Reply from 8.8.8.8 bytes = 43 time = 24.36 ms

```
import socket
import struct
import time
import os
def checksum(data):
    total = 0
    length = len(data)
for i in range(0, length-1, 2):
    total += (data[i] << 8) + data[i+1]
if length % 2
    total += data[-1] << 8
total = (total >> 16) + (total & 0xffff)
total += total >> 16
return (~total) & 0xffff
def ping(host):
    try:
    dest_addr = socket.gethostbyname(host)
except socket.gaierror:
    print("unable to resolve host")
return
sock.settimeout(2)
packet_id = os.getpid() & 0xFFFF
Sequence = 1
# ICMP Echo Request
header = struct.pack("!BBHHH", 8, 0, 0, packet_id,
                     Sequence)
data = b"computer Networks Ping"
packet = header + data
# Calculate Checksum
chksum = checksum(packet)
# Re-Create header with checksum
header = struct.pack("!BBHHH", 8, 0, chksum,
                     packet_id, sequence).
packet = header + data
start_time = time.time()
try:
sock_sendto(packet, (dest_addr, 1))
```

```
        sock.recvfrom(1024)
        end_time = time.time()
        rtt = (end_time - start_time) * 1000
        print("Reply from", dest_addr,
              ": bytes =", len(packet),
              "time = %.2f ms" % rtt)
    except socket.timeout:
        print("Request timed out")
finally:
    sock.close()

# Main program
host = input("Enter host name or IP address: ")
for i in range(4):
    ping(host)
    time.sleep(1)
```

Result:

Thus, a ping program was successfully implemented using python. The program sends ICMP Echo Request Packets to the Specified host, receives Echo Reply packets, and calculate the round-trip (RTT) to determine network connectivity.

Sample output

Enter hostname or IP address : 8.8.8.8

Reply from 8.8.8.8 : bytes = 43 time = 24.51 ms

Reply from 8.8.8.8 : bytes = 43 time = 23.87 ms

Reply from 8.8.8.8 : bytes = 43 time = 24.12 ms.

Reply from 8.8.8.8 : bytes = 43 min = 24.36 ms.